

The Enterprise GenAI Stack — corrected and extended

The “Full AI Stack Explained” graphic, re-researched for a regulated UK/EU asset manager · as of 9 October 2026

9 layers re-drawn

+8 enterprise controls

140 products assessed

39 of 80 original tiles out of date

1,255 sources

58 Strategic · 67 Tactical · 13 Experimental

Strategic (most are conditional)

Tactical (a stated niche or estate)

Experimental (pilot only)

Removed (unverifiable)

NEW not in the original

ACQUIRED owner changed 2025–26

RENAMED

AWS Strategic only where that cloud is primary

CONTROL PLANE — MISSING FROM THE ORIGINAL

firm-owned policy and one evidence store, around every call

C1 AI traffic gateway

model, tool (MCP) and agent (A2A) calls

Google Cloud

Apigee AI gateway

MCP support GA

Azure

Azure APIM AI gateway

GA policies only

agentgateway

MCP / A2A on Kubernetes

Kong AI Gateway

where Kong is the API standard

AWS

AgentCore Gateway

MCP gateway

Cloudflare AI Gateway

non-confidential work, BYOK

Portkey

Palo Alto Networks · Prisma AIRS

C2 Guardrails

policy and tests owned by the firm

AWS

Bedrock Guardrails

standalone API from the gateway

Google Cloud

Model Armor

strict residency

Azure

Azure AI Content Safety

Prompt Shields on documents

NeMo Guardrails

orchestrator · still 0.x

Llama Guard / Prompt Guard

one detector among several

ACQUIRED

Guardrails AI

Harvey-owned · hosted hub retired

C3 Privacy service (DLP / PII)

one API, six enforcement points

RENAMED

Presidio

behind a firm privacy-service API

Google Cloud

Sensitive Data Protection

Google's DLP service

RENAMED

Purview DSPM for AI

posture, not prompt-path DLP

Skyflow

vault tokenisation

Protegrity

existing customers

C4 Identity and authorisation

every agent a registered identity

OPA

policy decision point · CNCF

SPIFFE / SPIRE

workload identity · CNCF

Cedar

policy language · AgentCore Policy

Okta / Auth0 for AI Agents

where Okta holds the workforce

Entra Agent ID

where Entra holds the workforce

MCP Authorization

Anthropic-originated · alt: OAuth on OpenAPI

C5 Configuration of record

prompts, pins and manifests in Git

LangSmith Prompts

with LangSmith as L9 platform

ACQUIRED

Langfuse Prompts

with Langfuse as L9 platform

PromptLayer

framework-neutral registry

Prompts as code (Git)

the configuration of record

LaunchDarkly AgentControl

rollout between approved variants

C6 AI FinOps

cost per task, budgets fail closed

FinOps FOCUS

open billing data standard

Gateway cost attribution

budgets that fail closed

Vantage

reporting layer only

CloudZero

reporting layer only

ACQUIRED

Helicone

maintenance mode · migrate

C7 AI security

secrets, supply chain, sandboxing

ACQUIRED

HashiCorp Vault

IBM · agentic IAM · BUSL

Model & package scanning

safetensors by default

OpenSSF Model Signing

sign weights you produce

Prisma AIRS

Palo Alto estates

HiddenLayer

independent specialist

ACQUIRED

Check Point AI Guardrails

formerly Lakera Guard

C8 Model risk and governance

inventory, validation, evidence store

OpenLineage

+ firm GenAI facets

IBM watsonx.governance

IBM estates

Credo AI

policy-led programmes

Collibra AI Governance

where Collibra is the catalogue

ModelOp

after full due diligence

ValidMind

model-risk-led firms

EVALUATION & OBSERVABILITY PLANE

L9 joins C8 as one evidence plane with two owners

L9 Evaluation & observability plane

was: 9 · Evals & Observability

evidence for every layer, from day one

Repositioned from the bottom of the stack to a plane beside the controls. Phoenix and Arize are one owner now (Dynatrace).

NEW

MLflow GenAI

where an ML platform exists

ACQUIRED

Langfuse

ClickHouse-owned · self-host

LangSmith

LangGraph estates · BYOC, EU

Opik

Comet · Apache-2.0

ACQUIRED

Arize AX

Dynatrace-owned

DeepEval

CI metric library

ACQUIRED

Promptfoo

red-teaming · OpenAI deal announced

ACQUIRED

Arize Phoenix

Dynatrace-owned · ELv2

NEW

ACQUIRED

W&B Weave

CoreWeave-owned

Braintrust

eval-led teams

NEW

RENAMED

Datadog Agent Observability

Datadog APM estates

AGENT PLANE

L3 Orchestration: workflows & agents

was: 3 · Agent Frameworks

deterministic by default, durable, approval gates

Workflow vs agent is the key decision. Adds durable execution. AutoGen superseded; Agent Builder shuts 30 Nov 2026.

RENAMED

LangGraph

default · workflows + agents

NEW

Temporal

durable execution

Microsoft Agent Framework

GA · replaces AutoGen

Pydantic AI

typed agent steps

NEW

Google Cloud

Google ADK

on Agent Engine

NEW

AWS

Strands + AgentCore Runtime

framework + managed runtime

OpenAI Agents SDK

sandboxed sub-step · pre-1.0

CrewAI

use Flows for regulated work

RENAMED

LlamaIndex

retrieval toolkit

Vercel AI SDK

TypeScript app tier

RENAMED

Claude Agent SDK

Alpha · sandboxed only · alt: LangGraph

Mistral Agents API

Workflows in beta

L4 Tools & connectivity

was: 4 · Tools & Protocols

nothing reachable except through the governed gateway

Adds a mandatory tool-governance sub-layer. MCP and A2A moved to the Linux Foundation's AAIF.

A2A 1.0

cross-team delegation · signed cards

MCP

behind a governed gateway · alt: OpenAPI

NEW

AWS

AgentCore Gateway + Identity

managed tool governance

Browserbase

only where no API exists

E2B

microVM sandbox

Agent Skills

script-free, internal · alt: AGENTS.md

Exa

via egress proxy + DLP

ACQUIRED

Tavily

Nebius-owned

Composio

May 2026 token incident · self-host

KNOWLEDGE PLANE

L8 Ingestion & data preparation

was: 8 · Data Extraction

approved sources, ACLs and lineage on every chunk

Lineage, classification, DLP and ACL capture are built around replaceable parsers; runtime web access moves to L4.

Docling

default engine · LF AI & Data

RENAMED

Unstructured

ACL-aware connectors

NEW

Google Cloud

Google Document AI

processor region confirmed

Mistral OCR 4.1

pin the model ID

RENAMED

LlamaParse

parse and extract only

Firecrawl

AGPL server · cloud with ZDR

Reducto

hard documents

Apify

public data only

Crawl4AI

pre-1.0 · pilots

MinerU

licence thresholds apply

L7 Retrieval optimisation

was: 7 · Embeddings & Rerankers

pinned embed + rerank, hybrid fusion, eval gate

Embeddings and rerankers merged. EthicalAgents and Ragoos removed: they could not be verified.

Sentence Transformers

graphic said “SBERT” · self-host

Google Cloud

Gemini Embedding 2

EU endpoint excludes the UK

Cohere Embed 5 + Rerank

private deployment

OpenAI text-embedding-3

text baseline

ACQUIRED

Jina AI

Elastic-owned · CC-BY-NC weights

Qwen3 Embedding

self-host after review

ACQUIRED

Voyage AI

MongoDB-owned

RENAMED

NVIDIA NeMo Retriever

NVIDIA estates

EthicalAgents

removed · could not be verified

Ragoos

removed · could not be verified

L6 Retrieval & knowledge stores

was: 6 · Vector Databases

derived, entitlement-filtered, rebuildable index

“Vector database” is now a feature; hybrid search ships almost everywhere. Prefer the platform you already run.

Elasticsearch

hybrid + document-level security

Milvus / Zilliz

very large corpora

PostgreSQL + pgvector

where Postgres is standard

Qdrant

dedicated engine after a load test

MongoDB Vector Search

GA self-managed too

Pinecone

managed · BYOC

Weaviate

licence in transition

S3 Vectors

AWS cost tier · no BM25

turbopuffer

many tenants · BYOC only

Chroma

prototypes and harnesses

L5 Memory service (part of L6)

was: 5 · Memory

policy-gated writes, erasure by person

No longer a separate infrastructure layer. Built last. Zep CE deprecated, Letta pivoted, LangMem stalled.

NEW

AWS

AgentCore Memory

inside its own runtime

NEW

RENAMED

Google Cloud

Memory Bank

Agent Engine

Zep / Graphiti

Community Edition deprecated

Mem0

OSS behind a firm memory API

Cognee

in-estate only

Supermemory

v5 breaking change

Letta

now an agent harness

LangMem

no release since Oct 2025

MODEL PLANE

L2 Inference & model access

was: 2 · Inference & Access

in-region access; vLLM as the private exit route

Routing moves to the C1 gateway. TGI archived. Ollama and LM Studio are developer tools, not production.

PATTERN · NOT SCORED

Primary cloud's model service

Bedrock · Foundry · Gemini Enterprise Agent Platform, in region

vLLM

default engine · the exit route

Fireworks AI

once ISO certificates confirmed

SGLang

backup engine once CVE is fixed

Hugging Face Hub

governed weights · TGI archived

Together AI

EU dedicated, ZDR on

ACQUIRED

OpenRouter

behind the gateway · Stripe deal

Ollama

developer tier

Cerebras

low latency, non-confidential

LM Studio

desktops only · no service use

NEW

llm-d

CNCF sandbox · pilot

NEW

NVIDIA Dynamo

beta · pilot

L1 Foundation-model portfolio

was: 1 · LLMs

two unrelated vendors + small + open-weight, all pinned

“Gemma 2.9” does not exist (Gemma 4). “Q4” is Z.ai GLM. Mistral Medium 3.1 retired. xAI is now SpaceXAI.

OpenAI GPT-6

Astra · Sol · Luna · 6.1 Sol

Mistral

Medium 3.5 · Large 3 · 3.1 retired

Anthropic Claude

hyperscaler EU route · tier set by reader

Gemma 4

graphic said “Gemma 2.9”

Google Cloud

Google Gemini 3.x

pin versions · short lifetimes

ACQUIRED

Grok 4.7 (SpaceXAI)

via a hyperscaler only

DeepSeek V4

weights or in-tenant only

Z.ai GLM-5.3

graphic said “Q4”

RENAMED

Meta Muse / Llama

block the contributor tier

Qwen 3.8

self-host by policy

Kimi K3

custom licence

What changed since the original graphic

- 39 of 80 tiles were out of date: 9 acquired, 9 mispositioned, 8 renamed, 8 wrong version, 6 not verifiable, 4 duplicated, 3 superseded, 2 deprecated (some tiles carry several).
- Missing entirely: the control plane (C1–C8), durable execution, tool governance, and the hyperscaler agent stacks.
- Re-drawn: L2 split and routing moved to the gateway; workflows separated from agents; memory merged into the stores; embeddings and reranking merged; evaluation moved from the bottom of the stack to a plane beside the controls.
- Ownership moved: Dynatrace–Arize, ClickHouse–Langfuse, OpenAI–Promptfoo (announced), MongoDB–Voyage, Elastic–Jina, Nebius–Tavily, Stripe–OpenRouter (pending), SpaceX–xAI.

How to read it

- Tiers are for a regulated asset manager, scored on one rubric with regulated-FS weights. A Strategic tier carries a condition; the condition is the decision.
- Build order: control and evidence plane first, then models through the gateway, retrieval, workflows, tools — memory last.
- Disclosure: researched and drafted with an Anthropic model. Anthropic items were scored on the same rubric; their tiers were set by the reader, and an independent alternative is named for each.
- Personal research, not any firm's platform. Every tile is backed by sourced facts in the dataset and the product appendix.

Source: Enterprise GenAI Full-Stack Architecture review, October 2026 — 05_Data/products.json (tiers and flags), bibliography.xlsx (sources). Original graphic: “The Full AI Stack Explained” (October 2026). Product names are trademarks of their owners; no logos used.